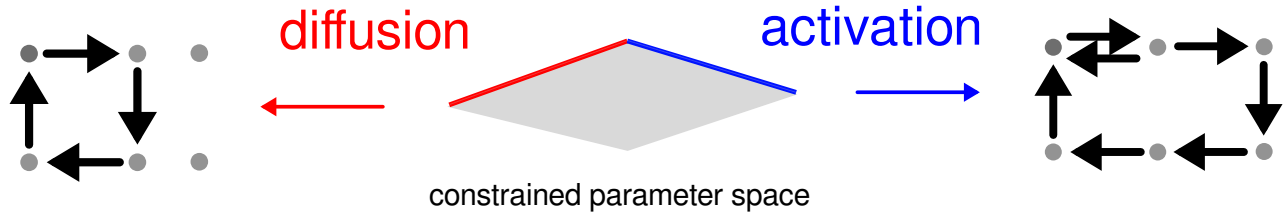The limiting constraint shapes the optimal architecture